



Customer Churn & Growth Analysis



Accelerating Strategic Growth and Customer Retention

A long-term strategic initiative focused on data-driven customer loyalty, personalized lifecycle engagement, and sustainable market expansion.

explore

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Find customers who are most likely to churn

```
SELECT
  *
FROM
  customer_churn
WHERE
  churn = 1 AND Payment_Delay < 5
  AND Support_Calls < 3;
```

Result Grid												
Filter Rows:												
Edit: Export/Import: Wrap Cell Content: Fetch rows:												
	CustomerID	Age	Gender	Tenure	Usage_Frequency	Support_Calls	Payment_Delay	Subscription_Type	Contract_Length	Total_Spend	Last_Interaction	Churn
	3042	18	Female	4	19	1	2	Premium	Monthly	529	4	1
	3046	54	Female	11	28	0	2	Premium	Annual	369	14	1
	3058	33	Female	53	20	0	4	Standard	Quarterly	475	5	1
	3061	47	Female	7	24	1	3	Premium	Quarterly	576	23	1
	3073	37	Female	28	18	1	3	Premium	Monthly	984	6	1
	3087	54	Male	55	4	2	3	Standard	Annual	879	28	1
	3113	46	Female	47	10	2	1	Basic	Monthly	655	16	1



Check whether customers with more support calls churn more

```
SELECT  
Support_Calls, COUNT(*) AS churn_customer_count  
FROM  
customer_churn  
WHERE  
churn = 1  
GROUP BY 1  
ORDER BY 2 DESC;
```

	Support_Calls	churn_customer_count
▶	10	23900
	7	23870
	9	23630
	5	23601
	8	23559
	6	23554
	4	22668
	3	21958
	0	21160
	1	21095
	2	21004

Summary for Stakeholders: The business must intervene before a customer reaches their 5th support call. Resolving issues early in the customer lifecycle is the single most effective way to reduce overall churn volumes based on current data.



Find which subscription type generates the highest revenue

```
SELECT  
Subscription_Type, SUM(Total_Spend) AS  
highest_revenue  
FROM  
customer_churn  
GROUP BY 1  
ORDER BY 2 DESC limit 1;
```

Result Grid			Filter Rows:	
	Subscription_Type	highest_revenue		
▶	Standard	94418293		

Summary for Stakeholders: > The Standard subscription tier is our cash cow, generating over 94.4M in total spend. Business strategy should heavily focus on maintaining the stability of this tier while experimenting with conversion paths to push these users toward higher-margin premium segments.

Analyze whether payment delay increases churn

```
SELECT  
Payment_Delay, COUNT(*) AS churn_customers  
FROM  
customer_churn  
WHERE  
churn = 1  
GROUP BY 1  
ORDER BY 1 DESC;
```

💡 **Summary for Stakeholders:** > Payment delays show a uniform distribution regarding churn volumes, meaning late payments are not a reliable metric for predicting customer exit. The business should standardize a firm 20-day cutoff policy to protect cash flow, while steering broader retention strategies toward product and customer service improvements rather than billing schedule

	Payment_Delay	churn_customers
	22	8454
	21	8670
	20	7996
	19	7785
	18	7861
	17	7847
	16	7995
	15	7988
	14	7938
	13	7929
	12	7940
	11	8065
	10	7869
	9	7871
	8	7959
	7	7888
	6	7915
	5	7806
	4	7852
	3	7930
	2	7853
	1	7793
	0	7889

Identify loyal customers using long tenure

```
SELECT
*
FROM
customer_churn
WHERE
churn = 0
ORDER BY tenure DESC limit 10;
```

CustomerID	Age	Gender	Tenure	Usage_Frequency	Support_Calls	Payment_Delay	Subscription_Type	Contract_Length	Total_Spend	Last_Interaction	Churn
8955	41	Male	60	28	1	5	Standard	Quarterly	705	10	0
7557	44	Female	60	19	2	1	Standard	Quarterly	675	11	0
1849	30	Female	60	4	1	13	Premium	Annual	747	9	0
879	40	Female	60	20	1	4	Basic	Quarterly	818	12	0
7240	37	Male	60	7	1	2	Premium	Quarterly	549	26	0
5545	43	Male	60	16	3	17	Premium	Annual	626	12	0
5451	48	Male	60	6	1	1	Basic	Quarterly	951	25	0
6037	39	Female	60	28	1	10	Basic	Annual	841	11	0
4552	39	Male	60	29	2	8	Standard	Quarterly	903	24	0
8020	33	Male	60	8	1	4	Standard	Quarterly	921	25	0
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

💡 Summary for Stakeholders: > Customers with a 60-month tenure are our lowest-maintenance, highest-value segment. Because their retention is highly correlated with low support friction, the strategic priority must be to proactively reward their loyalty and ensure that any rare support issue they encounter is handled with immediate, white-glove care.



Find high value customers with highest spending

```
SELECT  
CustomerID, SUM(Total_Spend) AS highest_spending  
FROM  
customer_churn  
GROUP BY 1  
ORDER BY 2 DESC  
LIMIT 1;
```

	CustomerID	highest_spending
▶	1890	1000

💡 **Summary for Stakeholders:** > Customer 1890 represents the absolute peak of individual lifetime value (LTV) at 1,000. The immediate business priority is to verify their current active status and provide immediate executive care to ensure this account remains highly satisfied and locked into our ecosystem.





Compare churn rate between male and female customers

```
SELECT  
Gender, COUNT(churn) AS churn_customer  
FROM  
customer_churn  
GROUP BY 1  
ORDER BY 2 DESC;  
select max(age) from customer_churn;
```

	Gender	churn_customer
▶	Male	250252
	Female	190580

⚠ A Quick Technical Note (The "Churn Rate" Trap): > While the query shows that more absolute numbers of males churned than females, this does not actually show the "churn rate." To find the true rate, you would need to divide these numbers by the total number of male and female customers in the database. If there are simply way more male customers overall, their churn rate might actually be lower or the same as females!





Find which age group uses the service the most

```
select case
  when age between 18 and 25 then '18-25'
  when age between 26 and 35 then '26-35'
  when age between 37 and 45 then '37-50'
  else '50+'
end as Age_group,
count(*) as customer_count from customer_churn
group by 1;
```

Result Grid			Filter Rows:
	Age_group	customer_count	
▶	26-35	104108	
	50+	155659	
	18-25	73683	
	37-50	107382	

The query categorizes customers into distinct age buckets using a CASE statement, counts the total number of records in each bucket (COUNT(*)), and groups the results together:

18-25

26-35

37-50 (defined in the query as ages between 37 and 45, but labeled '37-50')

50+ (captures all remaining ages, including age 36 which was skipped in the logic)

The Output Data

The result grid displays the customer distribution across these age groups:

50+ : 155,659 customers (Highest)

37-50 : 107,382 customers

26-35 : 104,108 customers

18-25 : 73,683 customers (Lowest)

Conclusion: The 50+ age group uses the service the mos





Calculate total company revenue

```
SELECT  
SUM(total_spend) AS Revenue  
FROM  
customer_churn;
```

Result Grid	
	Revenue
▶	278438024





Find average spending of premium subscription users

```
SELECT  
    Subscription_Type,  
    ROUND(AVG(total_spend), 2) AS average_spending  
FROM  
    customer_churn  
WHERE  
    Subscription_Type = 'Premium'  
GROUP BY 1  
ORDER BY 2 DESC;
```

	Subscription_Type	average_spending
▶	Premium	632.93

Table Used: customer_churn

- Filter Applied: Only rows where Subscription_Type = 'Premium'
- Calculation: Average spending of Premium users using AVG(total_spend)
- Rounding: Rounded to two decimal places with ROUND(..., 2)
- Grouping: Grouped by Subscription_Type
- Ordering: Ordered by average spending in descending order
-  Final Result
- Premium users spend an average of 632.93.



Check whether monthly contract users churn more

```
SELECT
Contract_Length, COUNT(*) AS churn_customer
FROM
customer_churn
WHERE
churn = 1
GROUP BY 1
ORDER BY 2 DESC;
```

	Contract_Length	churn_customer
▶	Monthly	87104
	Annual	81646
	Quarterly	81249

Action Plan (What the Company Should Do)

- **Incentivize Annual Plans:** Offer attractive discounts or free months (e.g., "Pay for 10 months, get 12") to push Monthly users into signing Annual contracts. This locks in long-term revenue.
- **Protect Premium Revenue:** Since Premium users spend the most, implement loyalty rewards, exclusive features, and priority support to ensure they stay happy and do not churn.
- **Introduce Quarterly Subscriptions:** Create a middle-ground 3-month or 6-month plan. This bridges the gap for users hesitant to buy a full year but prevents them from leaving after just one month.
- **Gather Exit Feedback:** Automate feedback surveys the moment a Monthly user cancels to understand why they are leaving and fix the underlying issues.
-



Find customers who never contacted support

```
SELECT  
CustomerID, Support_Calls  
FROM  
customer_churn  
WHERE  
Support_Calls = 0;
```

	CustomerID	Support_Calls
▶	13	0
	27	0
	30	0
	73	0
	76	0
	99	0
	104	0
	105	0

Action Plan (What the Company Should Do)

- **Proactive Engagement:** Reach out to these customers via automated emails or notifications to share tips, hidden features, or tutorials so they get more value out of the product.
- **Send Satisfaction Surveys:** Drop a quick NPS (Net Promoter Score) or feedback survey (e.g., "How are we doing?") to check if they are genuinely satisfied or silently struggling.
- **Monitor Usage Metrics:** Check their product usage activity. If their support calls are 0 and their usage activity is low, flag them as high-risk churn customers and offer a re-engagement incentive.





Calculate average tenure of churned customers

```
SELECT  
ROUND(AVG(tenure), 2) AS average_tenure  
FROM  
customer_churn  
WHERE  
Churn =1;
```

Result Grid		Filter
	average_tenure	
▶	30.47	





Find average payment delay of customers

```
SELECT  
ROUND(AVG(Payment_Delay), 2) AS  
avgerage_Payment_Delay  
FROM  
customer_churn;
```

Result Grid		Filter Rows:
	avgerage_Payment_Delay	
▶	12.97	





Analyze customer distribution by contract length

```
SELECT
Contract_Length, COUNT(*) customer_distribution
FROM
customer_churn
GROUP BY 1;
```

Contract_Length	customer_distribution
Annual	177198
Monthly	87104
Quarterly	176530

Executive Summary: Contract Length Optimization

The data shows a strong foundation in long-term contracts (Annual & Quarterly), while the Monthly segment is significantly underrepresented (less than half of the others). The strategy should focus on expanding the monthly pipeline, driving upgrades, and retaining the core base.

1. Expand the Monthly Customer Segment

Low-Risk Acquisition: Offer attractive trials or first-month discounts to capture customers hesitant about long-term commitments.

Highlight Flexibility: Emphasize "Cancel Anytime" in marketing campaigns to lower the barrier to entry.

2. Convert Monthly Users to Long-Term Plans

Targeted Upgrades: Send push notifications or email campaigns offering 20%–30% savings to monthly users who have been active for 3 to 6 months.

Premium Incentives: Add exclusive features or rewards to Annual/Quarterly plans that are unavailable on the monthly tier.

3. Retain the Core Base (Annual & Quarterly)

Proactive Renewals: Offer loyalty bonuses or discounts 1 month before their current contract expires.

Feedback & Engagement: Maintain high customer satisfaction through dedicated support and regular feedback loops, as these two segments represent the company's primary revenue.





Analyze relationship between usage frequency and spending

```
SELECT
    usage_frequency,
    ROUND(AVG(total_spend), 2) AS avg_spending,
    COUNT(*) AS total_customers
FROM
    customer_churn
GROUP BY usage_frequency
ORDER BY usage_frequency;
```

	usage_frequency	avg_spending	total_customers
▶	1	622.25	13797
	2	617.63	13633
	3	624.90	13843
	4	620.59	13549
	5	623.14	13716
	6	622.44	13746

- **High Churn Risk Identification:** Customers with low usage frequency (1 or 2) are paying the same high amount (~\$622) as heavy users. These low-engagement customers are at a high risk of churning because they are not getting enough daily value for their money.
- **Drive Engagement for Low-Frequency Users:** Implement targeted email or push campaigns for users in tiers 1–3 to increase their product adoption and safeguard renewal rates.
- **Review Pricing Strategy:** Since increased usage doesn't drive higher revenue, the company might consider introducing usage-based tiers or add-on features to monetize highly active users.





Collect risky customer data for churn prediction model

```
SELECT
customerid,
tenure,
usage_frequency,
support_calls,
payment_delay,
churn
FROM
customer_churn
WHERE
support_calls > 5 AND payment_delay > 10
AND usage_frequency < 3;
```

	customerid	tenure	usage_frequency	support_calls	payment_delay	churn
▶	40	53	2	10	13	1
	70	43	1	8	28	1
	236	1	1	7	18	1
	371	22	1	8	17	1
	435	44	2	6	26	1
	466	40	2	6	30	1
	563	11	2	7	16	1

- **The "Dead Zone" Profile:** Customers who combine low service engagement with high friction (frequent support issues and late payments) have a 100% churn rate in this sample.
- **Predictive Indicators:** This specific combination of features (support_calls > 5 AND payment_delay > 10 AND usage_frequency < 3) serves as an exact, high-confidence trigger for a churn prediction model.
- **Immediate Business Action:** The company must set up automated, real-time alerts. When an active customer starts hitting these exact thresholds, account managers or automated retention workflows must intervene before the contract is officially canceled.





Calculate average spending by age group

```
SELECT
  CASE
    WHEN age BETWEEN 18 AND 25 THEN '18-25'
    WHEN age BETWEEN 26 AND 35 THEN '26-35'
    WHEN age BETWEEN 37 AND 45 THEN '37-50'
    ELSE '50+'
  END AS Age_group,
  ROUND(AVG(total_spend), 2) AS averagae_spending
FROM
  customer_churn
GROUP BY 1;
```

	Age_group	averagae_spending
▶	26-35	645.60
	50+	603.22
	18-25	632.87
	37-50	658.37

- **Executive Summary: Average Spending by Age Group**
- The data reveals that average customer spending is relatively stable across all demographics, staying within a narrow range of \$603 to \$658. However, middle-aged adults are the highest spenders, while senior citizens spend the least.
- **Key Observations & Takeaways:**
- **Top Revenue Drivers:** The 37-50 age group leads with the highest average spending (\$658.37), closely followed by the 26-35 group (\$645.60). Working professionals are your most valuable segment.
- **Lowest Spending Segment:** The 50+ group has the lowest average spending (\$603.22).
- **Critical SQL/Data Gap Notice:** There is a logical gap in the SQL query logic (BETWEEN 37 AND 45 labeled as '37-50'). This means age 36 is completely missing from the data, and ages 46-49 are falling into the ELSE '50+' bucket. You should fix this query bug to ensure 100% accurate customer grouping.



Find which subscription plan has the lowest churn rate

```
SELECT
    Subscription_Type,
    COUNT(*) AS total_customer,
    SUM(churn) AS churn_customer,
    ROUND(SUM(churn) / COUNT(*) * 100, 2) AS churn_rate
FROM
    customer_churn
GROUP BY 1
```

	Subscription_Type	total_customer	churn_customer	churn_rate
▶	Standard	149128	83616	56.07
	Basic	143026	83210	58.18
	Premium	148678	83173	55.94

- **Executive Summary: Churn Rate by Subscription Type**
- The data shows that the company faces an alarmingly high overall churn rate (above 55%) across all subscription models. While the plans are evenly balanced in total customers (~143K to 149K), premium users show the highest retention.
- **Key Observations & Takeaways:**
- **Lowest Churn Plan:** The Premium tier has the lowest churn rate at 55.94%, closely followed by the Standard tier at 56.07%.
- **Highest Churn Plan:** The Basic tier has the highest churn rate at 58.18%. This suggests entry-level users find it easiest to abandon the service or see the least value in it.
- **Critical Business Emergency:** Regardless of which plan is "best," losing over half of your customer base across all tiers indicates a severe product, service, or onboarding issue that needs immediate investigation.



Find low spending but loyal customers for loyalty offers

```
SELECT  
*  
FROM  
customer_churn  
WHERE  
total_spend < (SELECT  
ROUND(AVG(total_spend), 2)  
FROM  
customer_churn)  
AND churn = 0  
ORDER BY tenure DESC;
```

Result Grid												
Filter Rows:												
Edit:												
Export/Import:												
Wrap Cell Content:												
Fetch rows:												
	CustomerID	Age	Gender	Tenure	Usage_Frequency	Support_Calls	Payment_Delay	Subscription_Type	Contract_Length	Total_Spend	Last_Interaction	Churn
▶	440868	35	Male	60	12	3	20	Premium	Quarterly	559	4	0
	441465	40	Female	60	14	2	16	Standard	Annual	554	6	0
	441590	47	Male	60	26	4	4	Standard	Annual	591	9	0
	441866	30	Male	60	30	0	13	Premium	Annual	507	19	0
	442344	44	30 Male	60	11	0	12	Standard	Annual	555	13	0
	442716	44	Male	60	7	2	3	Standard	Quarterly	507	8	0
	442725	23	Male	60	20	0	20	Standard	Quarterly	555	4	0
	442775	39	Male	60	24	0	3	Basic	Quarterly	543	2	0

customer_churn.8



Find which gender generates more revenue

```
SELECT  
gender, SUM(total_spend) AS revenue  
FROM  
customer_churn  
GROUP BY 1  
ORDER BY 2 DESC;
```

	gender	revenue
▶	Male	161542277
	Female	116895747

THANK YOU!

Questions? We are ready for the next phase of Plan
2030.

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